

Question Bank

UNIT-I

Section I: Multiple Choice Questions (MCQs)

1. What is the primary goal of data science?
 - (a) To program complex software applications
 - (b) To extract insights and knowledge from data
 - (c) To design and maintain computer networks
 - (d) To create visually appealing presentations
2. Which of the following is NOT a typical application of data science?
 - (a) Fraud detection in financial transactions
 - (b) Recommending products to customers online
 - (c) Designing user interfaces for mobile apps
 - (d) Predicting weather patterns
3. A data scientist is responsible for:
 - (a) Managing databases and servers
 - (b) Collecting and cleaning raw data
 - (c) Writing complex marketing copy
 - (d) Designing computer graphics
4. The term “Digital Universe” refers to:
 - a) Computer hardware
 - b) Collection of digital data created worldwide
 - c) Internet cables
 - d) Social media only
5. What is the first step in the data science life cycle?
 - (a) Model training and evaluation
 - (b) Data understanding and pre-processing
 - (c) Data visualization and communication
 - (d) Model deployment and monitoring
6. What is the first step in the data science life cycle?
 - (a) Model training and evaluation
 - (b) Data understanding and pre-processing
 - (c) Data visualization and communication
 - (d) Model deployment and monitoring
7. How is structured data different from unstructured data?
 - (a) Structured data is organized in tables, while unstructured data is not.
 - (b) Structured data is always numerical, while unstructured can be text or images.
 - (c) Structured data is more difficult to analyze than unstructured data.
 - (d) Structured data is created by humans, while unstructured data is generated by machines.
8. OSEMN framework was proposed by:
 - a) John Tukey
 - b) Hilary Mason

- c) Alan Turing
- d) Larry Page
- 9. The “O” in OSEMn stands for:
 - a) Organize
 - b) Obtain
 - c) Observe
 - d) Operate
- 10. Which stage of OSEMn involves cleaning data?
 - a) Model
 - b) Explore
 - c) Scrub
 - d) Interpret
- 11. Information Commons mainly supports:
 - a) Data sharing and collaboration
 - b) Mobile communication
 - c) Gaming applications
 - d) Hardware manufacturing
- 12. Structured data is commonly stored in:
 - a) Relational databases
 - b) Audio files
 - c) Videos
 - d) Images
- 13. Which of the following is an example of unstructured data?
 - a) Excel sheet
 - b) Database table
 - c) Social media posts
 - d) Attendance register
- 14. The “M” in OSEMn framework stands for:
 - a) Manage
 - b) Measure
 - c) Model
 - d) Merge
- 15. The Explore stage in OSEMn is mainly used for:
 - a) Prediction
 - b) Data visualization and analysis
 - c) Data deletion
 - d) Data storage

Section II – Short Answer Questions

1. Define Data Science.
2. What is meant by the Digital Universe?
3. List any four sources of data.
4. What is Information Commons?
5. Expand the term OSEMn.
6. What is structured data? Give one example.

7. What is unstructured data?
8. Differentiate between structured and unstructured data.
9. Mention any two applications of Data Science.
10. What is the role of the Scrub stage in OSEM?
11. Explain the importance of data visualization in Data Science.
12. What are IoT data sources?
13. State any two characteristics of Big Data.
14. What happens during the Explore stage of OSEM?
15. Define data cleaning.
16. Why is data important in Data Science projects?
17. Give examples of real-time data sources.
18. What is meant by data modeling?
19. What is the purpose of the Interpret stage in OSEM?
20. Mention any two challenges in Data Science projects.

Section III – Long Answer Questions

1. Explain the concept of Data Science and its importance in modern applications.
2. Discuss the Digital Universe and explain how digital data is growing rapidly.
3. Describe various sources of data with suitable examples.
4. Explain the concept of Information Commons and its significance.
5. Discuss the complete Data Science Project Life Cycle.
6. Explain the OSEM framework with neat diagram and examples.
7. Compare structured, semi-structured, and unstructured data.
8. Describe the role of data collection and preprocessing in Data Science projects.
9. Explain the importance of data exploration and visualization in the OSEM framework.
10. Discuss the applications of Data Science in healthcare, business, and education.
11. Explain the challenges faced in handling large-scale data in the Digital Universe.
12. Describe how Data Science integrates statistics, programming, and domain knowledge.
13. Explain each stage of the OSEM framework in detail.
14. Write short notes on:
 - a) Digital Universe
 - b) Information Commons
15. Explain the importance of data sources in building successful Data Science models.

UNIT-II

DATA PREPROCESSING

SECTION – I

1. Data preprocessing is mainly used to
 - a) Increase data size
 - b) Improve data quality
 - c) Delete all records
 - d) Create software
2. Which function is commonly used to read a CSV file in R?
 - a) read.table()
 - b) read.csv()
 - c) import.csv()
 - d) open.csv()
3. Selecting only required columns from a dataset is called
 - a) Sorting
 - b) Ranking
 - c) Selection
 - d) Grouping
4. Missing values in R are represented by
 - a) 0
 - b) NULL
 - c) NA
 - d) Empty
5. Which operation arranges data in ascending or descending order?
 - a) Filtering
 - b) Sorting
 - c) Grouping
 - d) Rearranging
6. Filtering data means
 - a) Removing unwanted records
 - b) Adding new columns
 - c) Ranking data
 - d) Grouping data
7. Which function is used to identify missing values in R?
 - a) is.na()
 - b) na.check()
 - c) missing()
 - d) check.na()
8. Grouping data is mainly used for
 - a) Data visualization
 - b) Organizing similar records together
 - c) Deleting rows
 - d) Rearranging columns

9. Ranking data is useful for
 - a) Ordering values based on priority
 - b) Removing missing values
 - c) Reading files
 - d) Creating tables
10. Rearranging data refers to
 - a) Changing structure or order of data
 - b) Removing duplicates
 - c) Reading datasets
 - d) Filtering records

SECTION – II

Short Answer Questions

1. Define data preprocessing.
2. What is meant by missing data?
3. Write the syntax to read a CSV file in R.
4. What is filtering of data?
5. Define sorting.
6. What is grouping in data preprocessing?
7. Mention any two preprocessing techniques.
8. What is ranking of data?
9. Why is data preprocessing important?
10. Differentiate between selecting and filtering data.

SECTION – III

Long Answer Questions

1. Explain the need and importance of data preprocessing.
2. Describe the steps involved in reading and selecting data in R with examples.
3. Explain different methods used for handling missing values.
4. Discuss filtering techniques used in data preprocessing.
5. Explain sorting and ranking of data with suitable examples.
6. Describe grouping and rearranging operations in data preprocessing.
7. Write a short note on data manipulation techniques.
8. Explain the process of preprocessing raw data before analysis.
9. Compare filtering, sorting, and grouping operations.
10. Explain various preprocessing operations in R with suitable examples.

UNIT III

CONCEPT LEARNING

SECTION I – MCQ TYPE QUESTIONS

1. In concept learning, a hypothesis is:
 - a) A database
 - b) A prediction model
 - c) A statement about the target concept
 - d) A clustering method
2. PAC learning stands for:
 - a) Probabilistic Accurate Concept
 - b) Probably Approximately Correct
 - c) Predictive Approximation Classifier
 - d) Partial Accurate Classification
3. The version space in Candidate Elimination Algorithm represents:
 - a) All possible datasets
 - b) Set of consistent hypotheses
 - c) Incorrect hypotheses only
 - d) Training examples
4. VC Dimension measures:
 - a) Training speed
 - b) Complexity of hypothesis space
 - c) Memory usage
 - d) Number of attributes
5. In Candidate Elimination Algorithm, S boundary represents:
 - a) Most general hypotheses
 - b) Most specific hypotheses
 - c) Incorrect hypotheses
 - d) Random hypotheses
6. Which learning model allows small error probability?
 - a) Exact learning
 - b) PAC learning
 - c) Unsupervised learning
 - d) Reinforcement learning
7. The G boundary in Candidate Elimination Algorithm contains:
 - a) Specific hypotheses
 - b) General hypotheses
 - c) Incorrect examples
 - d) Test data
8. A hypothesis consistent with all training examples belongs to:
 - a) Version space
 - b) Neural network

- c) Decision tree
- d) Clustering set
- 9. The Candidate Elimination Algorithm updates boundaries using:
 - a) Positive and negative examples
 - b) Random values
 - c) Hidden layers
 - d) Distance metrics
- 10. A higher VC dimension generally indicates:
 - a) Lower model complexity
 - b) Greater expressive power
 - c) Smaller dataset
 - d) Faster convergence

SECTION II – SHORT ANSWER QUESTIONS

1. Define concept learning.
2. What is a hypothesis in machine learning?
3. Explain the meaning of PAC learning.
4. Define VC Dimension.
5. What is version space?
6. Differentiate between S boundary and G boundary.
7. What is hypothesis elimination?
8. List the steps involved in Candidate Elimination Algorithm.
9. Why is VC dimension important in learning theory?
10. What are positive and negative training examples?

SECTION III – LONG ANSWER QUESTIONS

1. Explain the concept learning process with a suitable example.
2. Describe the formulation of hypothesis in machine learning.
3. Explain PAC learning in detail with advantages and limitations.
4. Discuss VC Dimension and its significance in concept learning.
5. Explain hypothesis elimination with an illustrative example.
6. Describe the Candidate Elimination Algorithm with step-by-step working.
7. Compare PAC learning and traditional exact learning approaches.
8. Explain version space representation and its importance in machine learning.
9. Demonstrate the working of Candidate Elimination Algorithm using a training dataset.
10. Discuss the role of hypothesis space in concept learning and machine learning models.

UNIT IV

Basics of R

SECTION I – MCQ TYPE QUESTIONS

1. Which of the following is used to assign a value in R?
 - a) <-
 - b) ==
 - c) =>
 - d) !=
2. Which data type is used to store TRUE or FALSE values in R?
 - a) Integer
 - b) Logical
 - c) Character
 - d) Complex
3. Which function is used to create a data frame in R?
 - a) matrix()
 - b) list()
 - c) data.frame()
 - d) vector()
4. Which loop is used when the number of iterations is known?
 - a) while loop
 - b) repeat loop
 - c) for loop
 - d) if loop
5. Which of the following is an ordered collection of elements in R?
 - a) Factor
 - b) Vector
 - c) Data frame
 - d) List
6. Scaling is mainly used to:
 - a) Remove rows
 - b) Convert text into numbers
 - c) Bring data into a common range
 - d) Delete missing values
7. Label Encoding converts:
 - a) Numbers into text
 - b) Categories into numeric labels
 - c) Rows into columns
 - d) Continuous data into factors
8. One Hot Encoding creates:
 - a) Multiple binary columns
 - b) One numeric column

- c) Duplicate rows
- d) Missing values
- 9. Which function is commonly used for dimensionality reduction in R?
 - a) scale()
 - b) prcomp()
 - c) cbind()
 - d) summary()
- 10. Which control structure is used for decision making in R?
 - a) if-else
 - b) for
 - c) while
 - d) repeat

SECTION II – SHORT ANSWER QUESTIONS

1. Define R programming language.
2. What are the different data types available in R?
3. Explain vectors in R with an example.
4. What is a data frame in R?
5. Differentiate between list and vector in R.
6. What are control structures in R?
7. What is feature engineering?
8. Explain the purpose of scaling in machine learning.
9. What is Label Encoding?
10. What is One Hot Encoding and why is it used?

SECTION III – LONG ANSWER QUESTIONS

1. Explain the basics of R programming with suitable examples.
2. Describe different data types and objects available in R.
3. Explain control structures in R with syntax and examples.
4. Discuss the creation and manipulation of data frames in R.
5. Explain feature engineering and its importance in data preprocessing.
6. Describe different scaling techniques used in data preprocessing.
7. Compare Label Encoding and One Hot Encoding with examples.
8. Explain dimensionality reduction and its advantages.
9. Write an R program to create a data frame and perform basic operations.
10. Explain the preprocessing steps involved before building a machine learning model in R.

UNIT V

MODEL FITTING IN R

SECTION I – MCQ TYPE QUESTIONS

1. Which function is commonly used in R to fit a linear regression model?
 - a) kmeans()
 - b) glm()
 - c) lm()
 - d) tree()
2. Logistic regression is mainly used for:
 - a) Clustering
 - b) Classification
 - c) Regression of continuous values only
 - d) Data cleaning
3. Which package in R is commonly used for Decision Tree models?
 - a) randomForest
 - b) e1071
 - c) rpart
 - d) cluster
4. Naïve Bayes classifier is based on:
 - a) Distance calculation
 - b) Probability theory
 - c) Tree structure
 - d) Neural networks
5. Which algorithm is best suited for finding hyperplanes for classification?
 - a) K-Means
 - b) SVM
 - c) Linear Regression
 - d) Hierarchical Clustering
6. Random Forest is an ensemble of:
 - a) Neural Networks
 - b) Regression Models
 - c) Decision Trees
 - d) Clustering Models
7. Which R function is used for K-Means clustering?
 - a) hclust()
 - b) glm()
 - c) kmeans()
 - d) predict()
8. Hierarchical clustering can be represented using:
 - a) Scatter plot
 - b) Histogram

- c) Dendrogram
- d) Pie chart
- 9. Which model is used when the dependent variable is categorical?
 - a) Linear Regression
 - b) Logistic Regression
 - c) K-Means
 - d) Hierarchical Clustering
- 10. Which package in R is widely used for SVM implementation?
 - a) e1071
 - b) caret
 - c) dplyr
 - d) ggplot2

SECTION II – SHORT ANSWER QUESTIONS

1. Define linear regression model in R.
2. What is the purpose of logistic regression?
3. Differentiate between regression and classification models.
4. What is a Decision Tree classifier?
5. Explain the concept of Naïve Bayes algorithm.
6. What is the role of kernel functions in SVM?
7. Define Random Forest model.
8. Explain the steps involved in K-Means clustering.
9. What is hierarchical clustering?
10. Write any two advantages of clustering techniques.

SECTION III – LONG ANSWER QUESTIONS

1. Explain Linear Regression in R with model fitting procedure and examples.
2. Discuss Logistic Regression and explain how it is implemented in R.
3. Describe the working of Decision Tree classification with suitable examples in R.
4. Explain the Naïve Bayes classification algorithm and its implementation in R.
5. Discuss Support Vector Machine (SVM) model and explain its advantages and limitations.
6. Explain the Random Forest algorithm and compare it with Decision Tree models.
7. Describe the K-Means clustering algorithm with steps and R implementation.
8. Explain Hierarchical Clustering and discuss agglomerative and divisive methods.
9. Compare Classification Models and Clustering Models used in R.
10. Explain the process of model fitting, training, testing, and evaluation in R for machine learning models.

UNIT VI

Visualization

SECTION I – MCQ TYPE QUESTIONS

1. Which type of graph is mainly used to show the distribution of numerical data?
 - a) Pie Chart
 - b) Histogram
 - c) Line Chart
 - d) Area Chart
2. A box plot is useful for identifying:
 - a) Network connections
 - b) Outliers
 - c) Database tables
 - d) Programming errors
3. Which visualization technique is best for showing the relationship between two numerical variables?
 - a) Scatter Plot
 - b) Heat Map
 - c) Bar Chart
 - d) Pie Chart
4. In a histogram, data is represented using:
 - a) Connected points
 - b) Circular sectors
 - c) Continuous bars
 - d) Icons
5. Heat maps are commonly used to represent:
 - a) Audio signals
 - b) Color-coded data intensity
 - c) Text formatting
 - d) Program execution
6. Tableau is mainly used for:
 - a) Operating system management
 - b) Data visualization and analytics
 - c) Hardware design
 - d) Compiler development
7. Which of the following indicates an outlier in a dataset?
 - a) A value close to the mean
 - b) A value significantly different from other observations
 - c) A repeated value
 - d) A missing value

8. Data balancing is mainly performed to:
 - a) Increase screen brightness
 - b) Equalize class distribution in datasets
 - c) Compress files
 - d) Improve internet speed
9. Which Tableau feature is used to create dashboards?
 - a) Worksheet
 - b) Story
 - c) Dashboard Pane
 - d) Data Grid
10. Which plot is most suitable for identifying correlation patterns?
 - a) Histogram
 - b) Box Plot
 - c) Scatter Plot
 - d) Heat Map

SECTION II – SHORT ANSWER QUESTIONS

1. Define data visualization.
2. What is a box plot? Mention its main components.
3. Differentiate between a histogram and a bar chart.
4. What is the purpose of a scatter plot?
5. Explain the use of heat maps in data analysis.
6. What are outliers in a dataset?
7. List any four features of Tableau.
8. What is data balancing?
9. Mention different methods used for outlier detection.
10. Explain the importance of visualization in data science.

SECTION III – LONG ANSWER QUESTIONS

1. Explain different types of data visualization techniques with suitable examples.
2. Describe the structure and working of a box plot. Explain how it helps in detecting outliers.
3. Discuss the construction and interpretation of histograms with real-world applications.
4. Explain scatter plots in detail and describe how they are useful in identifying relationships among variables.
5. What is a heat map? Explain its advantages, limitations, and applications.
6. Describe the features and architecture of Tableau. Explain how Tableau is used for data visualization.
7. Discuss various techniques used for outlier detection in data analysis.
8. Explain the concept of data balancing. Discuss different data balancing techniques used in machine learning.
9. Compare box plot, histogram, scatter plot, and heat map based on purpose, advantages, and limitations.

10. Create a detailed note on visualization tools and explain the role of Tableau in business intelligence and analytics.
- 11.

UNIT VII

PERFORMANCE EVALUATION IN R

SECTION – I

Multiple Choice Questions (MCQ)

1. Which of the following is used to measure the average squared difference between actual and predicted values?
 - a) Accuracy
 - b) Mean Squared Error
 - c) Recall
 - d) Precision
2. RMSE stands for:
 - a) Root Mean Standard Error
 - b) Relative Mean Square Error
 - c) Root Mean Squared Error
 - d) Residual Mean Square Error
3. Which evaluation metric is mainly used for classification problems?
 - a) MSE
 - b) RMSE
 - c) Accuracy
 - d) Variance
4. Precision is calculated as:
 - a) $TP / (TP + FN)$
 - b) $TP / (TP + FP)$
 - c) $TN / (TN + FP)$
 - d) $TN / (TN + FN)$
5. Recall is also known as:
 - a) Specificity
 - b) Sensitivity
 - c) Accuracy
 - d) Error Rate
6. Which metric combines Precision and Recall?
 - a) MSE
 - b) Accuracy
 - c) F1 Score
 - d) RMSE
7. In binary classification, True Positive means:
 - a) Incorrectly predicted negative class

- b) Correctly predicted positive class
- c) Correctly predicted negative class
- d) Incorrectly predicted positive class
- 8. Specificity measures:
 - a) Ability to identify positive cases correctly
 - b) Ability to identify negative cases correctly
 - c) Overall prediction error
 - d) Squared prediction loss
- 9. Which R function is commonly used to create a confusion matrix?
 - a) summary()
 - b) matrix()
 - c) confusionMatrix()
 - d) predict()
- 10. Lower RMSE value indicates:
 - a) Poor model performance
 - b) Better model performance
 - c) Higher classification error
 - d) Overfitting only

SECTION – II

Short Answer Questions

1. Define Loss Function in machine learning.
2. What is Mean Squared Error (MSE)?
3. Write the formula for Root Mean Squared Error (RMSE).
4. Differentiate between Accuracy and Precision.
5. Define Recall Score with suitable notation.
6. What is Binary Predictive Classification?
7. Explain the term Sensitivity.
8. What is Specificity in classification models?
9. Define F1 Score and mention its importance.
10. What is the role of a Confusion Matrix in model evaluation?

Section III – Long Answer Questions

1. Explain Loss Function and Error in detail with suitable examples in R.
2. Discuss Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) with formulas and R implementation.
3. Explain different model selection and evaluation criteria used in R.
4. Describe Accuracy, Precision, Recall, and F1 Score with confusion matrix representation.
5. Explain Binary Predictive Classification and its applications.
6. Discuss Sensitivity and Specificity in detail with suitable examples.
7. Compare classification evaluation metrics and explain when each metric should be preferred.
8. Write an R program to calculate Accuracy, Precision, Recall, and F1 Score for a dataset.

9. Explain the importance of performance evaluation in machine learning models.
10. Describe the complete process of evaluating a predictive model in R using suitable metrics.